

Navigational Rules of the Road

INTERNATIONAL / INLAND

Based on 33 CFR: Navigation and Navigable Waters PART 83—RULES

**What every boater needs to know...
Explained so every boater can understand**

Updated to include August 2014 changes

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SECTION 1: NAVIGATION RULES OF THE ROAD

Learning Objective:

The student will have a working understanding of both the International and Inland Navigation Rules.

These rules will *only make sense* if you place yourself in a boat 100 feet long or larger. Most small boats have exceptions.

The purpose of these Rules is to *avoid collisions!* As we review each one, keep that point in mind – avoid collision with vessels and other objects.

Navigation Rules consist of International and Inland Rules separated by the COLREGS Demarcation Lines. They delineate waters upon which mariners shall comply with the Inland and International Rules. The Demarcation lines are found at mouths of harbors along the coasts.



International Rules formalized in the Convention on the International Regulations for Preventing Collisions at Sea. In 1972 they determined 72 COLREGS to be applicable on waters outside of established navigational lines of demarcation.

Inland Rules replaced the old Inland Rules, Western Rivers Rules, Great Lakes Rules, their respective pilot and interpretive rules, parts of the Motorboat Act of 1940 and Inland Navigational Rules Act of 1980.

There are 38 Rules plus five Annexes. With the exception of Annex V, which is only Inland Rules, the International and Inland Rules and Annexes are very similar in both content and format.

Grouping of the Rules as we will discuss them: Rules 1 to 3 - General; Rules 4 to 10 - Conditions of Visibility; Rules 11 to 18 - Vessel conduct in sight of one another; Rule 19 - Conduct of vessels in restricted visibility; Rules 20 to 31 - Application lights and shapes; Rules 32 to 37 - Sound and light signals; Rules 38 - Exemptions; and Annex 1 to 5 (we will talk about No. 5).

Rule Measurements

7 m = 23 ft; 12 m = 39.4 ft; 20 m = 65.6 ft; 50 m = 164 ft; 100 m = 328 ft

RULE 1: Application of Rules

Applies to *all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels.* These Rules apply to all vessels upon the inland waters of the United States, and to vessels of the United States on the Canadian waters of the Great Lakes to the extent that there is no conflict with Canadian law. The regulations in this subchapter (Subchapter E, 33 CFR parts 83 through 90) have preemptive effect over State or local regulation within the same field.

Nothing in these Rules shall interfere with the operation of any special rules made by the Secretary of the Navy related to warships or fishing fleets.

Whenever the Secretary of Homeland Security may issue a certificate of alternative compliance for a vessel or class of vessels specifying the closest possible compliance with these Rules.

The operator of each self-propelled vessel 12 meters or more in length shall carry, on board and maintain for *ready reference*, a copy of these Rules.

RULE 2: Responsibility

Nothing in these Rules shall exonerate any vessel, or the owner, master, or crew thereof, from the consequences of any neglect to comply with these Rules or of the *neglect of any precaution which may be required by the ordinary practice of seamen*, or by the special circumstances of the case.

This is a very powerful and important point. As the person in charge, you are responsible for actions of your vessel and staff, not just as prescribed by rule but also by common sense.

“Complying with these rules, due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessels involved which may make a departure from these Rules necessary to avoid immediate danger.” Any special circumstances means more than two vessels who have a risk of collision.

You are going to take whatever actions are needed to avoid a collision. Using common sense, we depart from the rules to avoid *“immediate danger.”*

RULE 3: General Definitions

Here are several definitions you will need to better understand the rules.

Vessel includes every description of water craft, including nondisplacement craft and seaplanes, used or capable of being used as a means of transportation on water; *if it floats, it's a boat.*

Power-driven vessel means any vessel propelled by machinery.

Sailing vessel means any vessel under sail provided “that propelling machinery, if fitted, is not being used.”

Vessel engaged in fishing means any vessel fishing with nets, lines, trawls, or other fishing apparatus which *restricts maneuverability*, but does not include trolling lines or other fishing apparatus which do not restrict maneuverability.

Seaplane includes any aircraft designed to maneuver on water.

Vessel not under command is a vessel through some exceptional circumstance that is unable to maneuver as required by these Rules and is unable to keep out of the way of another vessel.

Vessel restricted in her ability to maneuver is a vessel which, from the nature of her work, is restricted in her ability to maneuver as required by these Rules and is unable to keep out of the way of another vessel. *(Examples include: dredging, surveying, or underwater operations; transferring persons, provisions, or cargo underway; launching/recovery of aircraft; mine clearance operations; or, servicing navigation marker.)*

Vessels shall be deemed to be in sight of one another only when one can be observed visually from the other.

Restricted visibility means any condition in which visibility is restricted by fog, mist, falling snow, heavy rainstorms, sandstorms, or any other similar causes. Darkness is **NOT** restricted visibility. You can still see the other boat's lights.

Underway means that a vessel is not at anchor, or made fast to the shore, or aground.

Western Rivers means “the Mississippi River, its tributaries, South Pass, and Southwest Pass, to the navigational demarcation lines . . .”

Great Lakes means “the Great Lakes and their connecting and tributary waters, including a portion of the Calumet River, the Chicago River, and the Saint Lawrence River . . .”

Inland Waters are “navigable waters of the U.S. shoreward of the demarcation lines...harbors, rivers, other inland waters of the U.S., and the waters of the Great Lakes on the U.S. side of the International Boundary.”

Vessel constrained by her draft means a power-driven vessel which, because of her draft in relation to the available depth and width of navigable water, is severely restricted in her ability to deviate from the course she is following. (*International Rules Only*)

RULE 4: Application

Rules 4 to 10 apply to any condition “of visibility.” The key word is *any*, day, night, restricted, etc.

RULE 5: Look Out

Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

Key Items:

- Sight and hearing
- At all times
- Full appraisal
- Means appropriate prevailing circumstances/conditions

So who is your look-out when you are teaching sailing or landing a client’s fish? Remember, it is “by all available means appropriate in the prevailing circumstances and conditions...” What about that radar unit no one knows how to operate?

Two Hungarian tourists killed on July 7, 2010, when a disabled amphibious Duck boat was struck by a 250-foot barge being towed by a tugboat. Probable cause was the failure of the tugboat’s mate to maintain proper lookout. He was operating from lower wheelhouse, where visibility was limited, and was distracted by cell phone usage and use of a company-issued laptop. Wheelhouse distraction!

RULE 6: Safe Speed

Every vessel shall at all times proceed at a safe speed so that she can take *proper and effective action* to avoid collision, and be *stopped within a distance* appropriate to the prevailing circumstances and conditions.

The following items determine what is a safe speed when complying with these two requirements.

Safe speed is determined by the following factors, plus any other needed to avoid a collision:

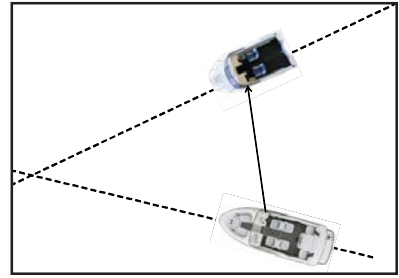
- Visibility
- Traffic density
- Maneuverability - stopping distance/turning ability in “the prevailing conditions”
- Background light from shore lights or back scatter of own lights
- Wind, sea, and current
- Navigational hazards
- Draft related to depth of water

RULE 7: Risk Of Collision

Proper use shall be made of radar equipment, if fitted and operational, including long-range scanning to obtain early warning of risk of collision and radar plotting or equivalent systematic observation.

Assumptions shall not be made on the basis of scanty information, especially scanty radar information.

Risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change. Meaning, if the compass bearing does not change but the range is decreasing, it is getting closer to you and a risk of collision exists.



Please Note: Such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow, or when approaching a vessel at close range. Again, if not sure, assume there is a risk of collision and act accordingly.

RULE 8: Action to Avoid Collision

Any action taken to avoid collision shall be taken in accordance with the Rules of this subpart (Rules 4-19) and shall, if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship.

Any alteration of course or speed to avoid collision shall be large enough to be **readily apparent** to another vessel observing visually or by radar; a succession of small alterations of course or speed should be avoided. When making speed or direction change, make them big enough so others can see it (large and apparent).

If there is sufficient sea room, alteration of course alone may be the most effective action to avoid a close-quarters situation, provided it is made in good time, is substantial, and does not result in another close-quarters situation.

Action taken to avoid collision with another vessel shall be such as to result in passing at a safe distance. The effectiveness of the action shall be carefully checked until the other vessel is finally past and clear.

If necessary, to avoid collision or allow more time to assess the situation, a vessel shall slacken her speed or take all way off by stopping or reversing her means of propulsion.

Need the time to assess your options? *Normally, you can not go wrong by slowing down to bear steerage to check things out.*

RULE 9: Narrow Channels

A vessel proceeding along a narrow channel or fairway shall keep as near to the outer limit of the channel or fairway to her starboard side as is safe/practical. A power-driven vessel, operating in narrow channels or fairways on Great Lakes, Western Rivers, or waters specified by the Secretary and proceeding down bound with a following current shall:

- have the right-of-way over an upbound vessel,
- propose the manner and place of passage, and
- initiate maneuvering signals as appropriate.

The vessel proceeding upbound against the current shall hold as necessary to permit safe passing. This is the only boat, by rule, that has the **“right of way.”** All the rest are stand-on, give-way, or not status.

There are several actions small vessels must take so larger vessels which must stay in the channel can operate without risk.

- A vessel less than 20 meters (65 ft) in length or a sailing vessel shall not impede the passage of a vessel that can safely navigate only within a narrow channel or fairway.
- A vessel engaged in fishing (commercial fishing) shall not impede the passage of vessel navigating within a narrow channel.
- A vessel shall not cross a narrow channel if crossing impedes the passage of a vessel needing to navigate within the channel.
- Any vessel shall, if the circumstances of the case admit, avoid anchoring in a narrow channel.

Vessel nearing a bend or an area—where other vessels may be obscured by an obstruction—shall navigate with particular alertness and caution and shall sound the appropriate signal (one prolong blast). *They can not see you, so they let you know by sounding the one prolong blast.*

RULE 10: Traffic Separation Schemes

Traffic separation schemes are similar to our lanes on a highway. They are used to safely move traffic in crowded areas. Instead of painted lines, there are buoys and charts outlining the lanes. Using traffic separation schemes does not relieve any vessel of her obligation under any other Rule.

Small boats stay out of these areas. If you must cross, first call the traffic control officer.

A vessel using a traffic separation scheme shall:

- Proceed in the direction of traffic flow for that lane.
- Keep clear of a traffic separation line/zone.
- Join/leave a traffic lane at the termination.
- Join/leave from side at smallest angle to traffic flow.
- Avoid crossing traffic lanes; if doing so, shall cross at right angles to the traffic flow.
- Avoid anchoring in a traffic separation scheme or near its terminations.

RULE 11: These rules apply when I can see you with my eyes (Rules 11-18)

RULE 12: Sailing Vessels

When two sailing vessels are approaching each other and involve risk of collision, one of them shall keep out of the way of the other as follows:

- When each has the wind on a different side, the vessel which has the wind on the port side (port tack) shall keep out of the way of the other.
- When both have the wind on the same side, the one to windward shall keep out of the way of the one to leeward.
- If a vessel with the wind on the port side sees a vessel to windward and cannot determine its tack, keep out of the way of the other.

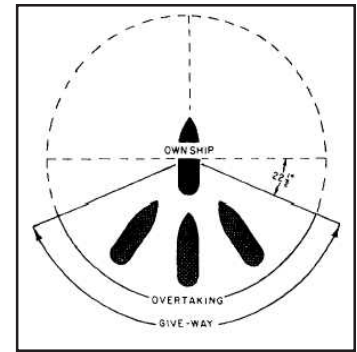
RULE 13: Overtaking

Overtaking is a passing situation. Any vessel overtaking any other shall keep out of the way of the vessel being overtaken. Simple and to the point. ***Note: This is for any vessel, not just power.***

A vessel is overtaking when coming upon a vessel from a direction more than 22.5 degrees abaft her beam, or at night you only see the sternlight.

Any subsequent alteration of the bearing between the two vessels shall not make the overtaking vessel a crossing vessel or relieve her of the duty of keeping clear of the overtaken vessel until she is finally past and clear. Your status does not change until you have completed the passing maneuver.

Memory Aid: *P-O-W have no rights.* (Port, Overtaking, Windward are give-way vessels.)



RULE 14: Head-On Situation

Unless otherwise agreed upon, when two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision, each shall alter her course to starboard so that each shall pass on the port side of the other.

What do you do if you meet a car on a road? You move to the starboard (right) to allow the other vehicle to pass on your port (left) side.

Head-on situation exists when a vessel sees the other ahead or nearly ahead. By night she could see the masthead lights of the other in a line or nearly in a line or both sidelights. By day she observes the corresponding aspect of the other vessel. If in any doubt, assume that it does exist and act accordingly.

The Rule concerning a power-driven vessel on the Western Rivers, proceeding downbound with a following current takes precedent over this rule.

RULE 15: Crossing Situation

When two power-driven vessels are crossing, so as to involve risk of collision, the vessel with the other on "her starboard side shall keep out of the way, and shall, if the circumstances of the case admit, ***avoid crossing ahead of the other vessel.***" Again, the Western River rule takes precedent in reference to downbound.



They see your green starboard light which says "go." You see their red port light which says "stop." Give way to vessels on your starboard side.

RULE 16: Action By Give-Way Vessel

Every vessel which is directed to keep out of the way of another vessel shall... take early and substantial action to keep well clear.

Give way vessels give way. Simple and to the point. When taking actions, the give way vessel must, if possible, avoid crossing ahead of the other vessel.

RULE 17: Action By Stand-on Vessel

Where one of two vessels is to keep out of the way, the other *shall keep her course and speed.*

"The stand-on vessel may...take action to avoid collision by her maneuver alone, as soon as it becomes apparent to her that the vessel ...is not taking appropriate action in compliance with these Rules."

When the stand-on vessel finds herself so close that collision cannot be avoided by the action of the give-way vessel alone, *she shall take such action as will best aid to avoid collision.*

She shall NOT, if possible, turn to port for a vessel on her own port side.

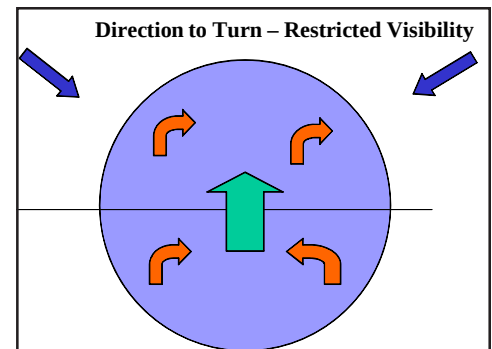
RULE 18: Responsibility Between Vessels

Except in a narrow channel, traffic separation scheme, or overtaking, vessels below stay out of the way of all vessels above it. This rule is easy to misunderstand. It is only when you are crossing or head-on and NOT overtaking. Also referred to as open water priority. ***You give way to the vessel(s) above your's.***

- Vessel not under command.
- Vessel restricted in her ability to maneuver.
- Vessel constrained by her draft (*International Rules only*).
- Vessel engaged in fishing/trawling, (not trolling).
- Sailing vessel, undersail and NOT motoring.
- Power driven vessel, underway (including towing, pilot, and safety.)
- Seaplane on the water.
- Wing in ground craft when taking off, landing, and in flight near the surface, keep well clear of all other vessels and avoid impeding their navigation; and on water comply with Rules as a power driven vessel.

Memory Aid for this list of privileged vessels is:

- New Not Under Command
- Reels Restricted in Ability to Maneuver
- Catch Constrained by Draft (*International Rules Only*)
- Fish Fishing Vessel or Trawler, Commercial
- So Sail Only (Not under power)
- Purchase Power-Driven
- Some Seaplane



RULE 19: Conduct in Restricted Visibility

Applies to vessels not in sight of one another when navigating ***in or near*** an area of restricted visibility.

- Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility.
- A power-driven vessel shall have her engines ready for immediate maneuver.
- Every vessel shall have due regard to the prevailing circumstances and conditions of restricted visibility when complying with Rules 4 through 10.
- (*If you detect by radar alone.*) The presence of another vessel shall determine if a close-quarters situation is developing or risk of collision exists. If so, she shall take avoiding action in ample time, provided action of alteration of course, so far as possible, shall avoid:
 - an alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken.
 - an alteration of course towards a vessel abeam or abaft the beam.

Without radar we will do the following (*except when a risk of collision does not exist*):

- Every vessel which hears (*apparently forward of her beam*) the fog signal of another vessel, or which cannot avoid a close quarters situation with another vessel forward of her beam, *shall reduce her speed to the minimum at which she can be kept on course. Bare steerage.*
- She shall, if necessary, take all her way off and, in any event, navigate with extreme caution until danger of collision is over. However, with no water flowing over the rudder, it's hard to turn to avoid another ship.

Quick Review: You must be able to see these vessels with your eyes for this rule to apply.

Sailboats

Different tack - Port, give way

Same tack - Windward, give way

Power Driven

Over-taking - Always give way

Head-on - No stand on, go to starboard

Crossing - Give way to starboard

Give way - Stay out of the way

Stand on - Maintain course and speed but "may/shall take action, if needed"

Open Water Priority

New - Not under command

Reels - Restricted in ability to maneuver

Catch - Constrained by draft (International only)

Fish - Fish/trawler

So - Sail only (not under power)

Purchase - Power driven

Some - Seaplane

Restricted Visibility - Bare steerage and proceed with caution

RULE 20: Application Lights and Shapes

Required lights must be displayed from sunset to sunrise **or**, from sunrise to sunset in restricted visibility, **or**, lights may be exhibited in all other circumstances when it is deemed necessary.

- No other lights may be displayed unless they can not be confused with the navigation lights or interfere with their function.
- Other lights "must not" interfere with the keeping of a proper look-out. In other words, do not destroy the look-out's night vision.
- Day shapes shall be complied with during the day.
- A vessel's navigation lights and shapes may be lowered if necessary to pass under a bridge.

RULE 21: Definitions Of Lights

Each navigation light has a specific purpose. They define the vessels type, stand-on/give-way status, side you see, and direction of travel. All of these are important for safe navigation.

Masthead light is placed over the vessels centerline showing an unbroken arc of light of **225 degrees** over the horizon. It faces forward. It is visible 22.5 degrees abaft the beam on either side of the vessel. It is always a white light. It can not be seen with sternlight.

Sidelights are a green light on the starboard side and red light on the port side, each showing an unbroken arc of light 112.5 degrees over the horizon. They can be seen from straight ahead to 22.5 degrees abaft the beam on its respective side. If the vessel is less than 20 meters, sidelights may be combined in one lantern carried on the vessel's centerline.

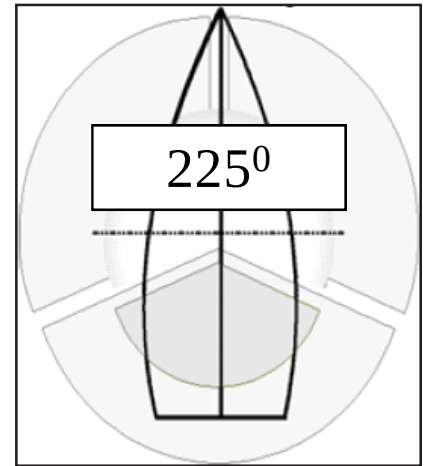
Sternlight is a white light placed at the stern. It shows an unbroken light over an arc of the horizon of **135 degrees**. If you see the sternlight, you should not see the side or masthead lights.

Towing light is a yellow light having the same characteristics as the sternlight but is not a sternlight.

All-round light means a light showing an unbroken light over an arc of the horizon of 360 degrees.

Flashing light is a light flashing at regular intervals of 120 flashes or more per minute.

Special flashing light means a yellow light flashing at regular intervals at a frequency of 50 to 70 flashes per minute. (*Inland Rules Only*) It is placed as far forward and on centerline as practicable of tow. It shows an unbroken light arc 180 degrees to 225 degrees.



RULE 22: Visibility Of Lights

Memory Aid: “*Big boats bright lights; Small boats dim lights.*”

- Vessel of 50 meters or more in length: masthead light, 6 miles; other lights, 3 miles; special flashing light, 2 miles.
- Vessel of 12 meters < 50 meters in length: masthead light, 5 miles; vessel < 20 meters, 3 miles; others, 2 miles.
- Vessel < 12 meters in length: masthead light, 2 miles; others, 2 miles; sidelight, 1 mile.
- In an inconspicuous, partly submerged vessel or object being towed: a white all-round light, 3 miles.

RULE 23: Power-Driven Vessels Underway

Lights on vessels can be a little confusing at times. Lights tell us two things: 1) the direction of travel, and 2) status. However, if we start with **standard light configuration** for a power-driven vessel, we have a solid understanding of lights. For the remaining vessels we change the masthead lights or add other lights where they can best be seen. Lights of different types of vessels show us their status in relationship to a power-driven vessel: stand-on vs. give-way vs. neither; underway vs. anchored or aground; or any dangers when passing vessels.

Standard light configuration starts with a power-driven vessel. More lights are added as the size and status of the vessel change.

A power-driven vessel underway under 50 meters exhibits:

- masthead light forward,
- sidelights, and
- sternlight.

A power-driven vessel of less than 12 meters may exhibit:

- all-round white light and sidelights (*Inland Rules Only*).

Note: The masthead light that is abaft and higher appears lower when you're up close to the vessel. Side lights can be just about anywhere. On large ships, they are on the main super structure.

- A power-driven vessel operating on the Great Lakes may carry an all-round white light, in lieu of the second masthead light and sternlight, and carry it in the position of the second masthead light.
- An air-cushioned vessel operating in the nondisplacement mode must display the normal lights for her length, plus an all-round flashing yellow light where it can best be seen.

- (*International Rules Only*) A power-driven vessel of less than 7 meters in length and whose maximum speed does not exceed 7 knots may exhibit an all-round white light and shall, if practicable, also exhibit sidelights.

Remember! Our small boats are an exception to the rules.

A power-driven vessel underway, 50 meters or longer in length, add a second masthead light abaft and higher of the forward one. Why? *It tells you which way the vessel is traveling and there is a lot of space between front and back.*

RULE 24: Towing and Pushing (Towing Astern)

A power-driven vessel, when towing, has no special status. There are a number of variations we will cover on the next several pages.

(*Both Inland and International Rules*) A power-driven vessel, when towing astern, shall exhibit:

- two masthead lights in a vertical line,
- sidelights,
- sternlight,
- towing light in a vertical line above the sternlight, and
- a third masthead light is added if tow, from stern of tug to after end of tow, exceeds 200 meters.
- And, when tow length exceeds 200 meters, a diamond shape is added where it can best be seen.

Towing light over the sternlight tells us there is a danger behind this boat. If hit, both the tow line and tow can cause serious problems.

Memory Aid: “Yellow over white; we are NOT alone tonight.”

RULE 24: Towing and Pushing (Towed Vessel)

Vessels or objects being towed will display: sidelights; sternlight; and (if tow is over 200 meters) a diamond shape where it can best be seen. **Note:** *Each boat* that is a tow must be lighted this way.

What lighting changes are made towing astern when you cross the COLREGs line? *None.*

Composite Vessel is the exception. *A composite vessel is lit as a power-driven vessel.* It can not show the light of a towing vessel.

RULE 24: Towing and Pushing (Pushing/Alongside)

(*International Rules*) A power-driven vessel pushing ahead or alongside (also called a hip tow) will display:

- two masthead lights in a vertical line
- sidelights
- sternlight
- no towing lights

The only added light to the standard configuration is a second masthead light in a vertical line.

(*Inland Rules Only*) A power-driven vessel when pushing ahead or towing alongside will display:

- two masthead lights in a vertical line,
- sidelights, and
- two towing lights in a vertical line, and NO sternlight.

Memory Aid: “Yellow over yellow, you are a pushy fellow.” This light arrangement will **NOT** be seen Internationally!

When a push tow crosses the COLREGs line from International to Inland Rules, we turn off the stern light and add two towing lights. On the front of the barge we add the special flashing light.

RULE 24: Towing and Pushing (Other)

With multiple vessels being towed alongside in a group, lighted as one vessel, and **Towed Alongside:**

- sternlight, if outside,
- sidelights, and
- special flashing light.

With multiple vessels being towed by pushing in a group, lighted as one vessel, and **Being Pushed Ahead:**

- forward sidelights,
- special flashing light so tow can be seen from the back, and
- NO sternlight on barge in front of tug.

With vessels on **Both Sides of the Towing Vessel:**

- sternlight on outboard vessel on each side,
- single set of sidelights, far forward/outboard as possible, and
- single special flashing light.

Inconspicuous, partly-submerged vessel/object being towed has special lighting needs; shape/size may vary:

- one all-round white light at, or near, each end,
- (If 25 meters or more in breadth), four all-round white lights to mark its length and breadth,
- (If 100 meters in length), additional all-round white lights so the distance between the lights is greater than 100 meters, and/or
- diamond shape at, or near, the aftermost extremity of the tow. Towing vessel may direct a searchlight in direction of tow to indicate its presence. (*Inland Only*)

On **Western Rivers above the Huey P. Long Bridge on the Mississippi River**, a power-driven vessel pushing ahead, or towing alongside, need only exhibit:

- sidelights;
- two towing lights in vertical line; and
- no masthead lights. (Needless to say, *Inland Rules Only*)

Note: When it is impracticable for a vessel not normally engaged in towing operations to display the required lights, you do not have to exhibit such lights. However, all possible measures shall be taken to indicate the relationship between the towing and assisted vessel; i.e., searchlight on the tow.

Towing Review

When, and where, will we see special flashing lights? *On the front of a barge (Inland Rules) when on the hip or in front.* The **ONLY yellow light** when towing in International Rules areas is on astern tow!

Memory Aids:

"Yellow over white, we are not alone tonight." (Something is behind the tug, a stern tow.)

"Yellow over yellow, you are a pushy fellow." (Hip or front, *Inland Rules Only*)

To the standard configuration, we added two masthead lights and a tow light is required.

RULE 25: Underway Vessels Sailing/Under Oars

Sailing vessel shall exhibit:

- sidelights
- sternlight, and
- may exhibit (at, or near, the top of the mast, where they can best be seen) two all-round lights in a vertical line, the upper being red and the lower green.

Memory Aid: *"Red over green, a sailing machine."*

Light Change from standard configuration? *No masthead light, "as it is not powered," with "optional light at the top."*

Dayshape: If you are undersail, and also propelled by machinery, you are considered a power-driven vessel with sails up. You must exhibit forward (where it can best be seen) a conical shape (apex downward) unless you are less than 12 meters in length. What dayshape for a sailing vessel? *None, when it meets the definition of sailing (with sails up and no motor propelling the vessel).*

Sailing vessel < 7 meters (23 ft) in length, **or a vessel under oars**, shall exhibit (if practicable) the side lights and stern light, but if she does not, she shall exhibit an all-round white light or have ready at hand an electric torch or lighted lantern showing a white light which shall be exhibited in sufficient time to prevent collision.

These are *Federal and International Rules*, so it is best to check with your State regulations as well before you leave in a sail or oar vessel without lights at night.

RULE 26: Fishing Vessels

Vessels engaged in fishing (other than trawling) and carrying fishing gear are limited/restricted in their ability to comply with the rules, whether underway or at anchor. They shall exhibit:

- two all-round lights in a vertical line, the upper being red and the lower white, and
- when ***making way***, exhibit sidelights and a sternlight.

Note: Trolling is NOT fishing, by this definition.

Memory Aid: *"Red over white, lines are in sight."*

Vessels engaged in trawling, by dragging a dredge net or other fishing appliance, shall exhibit:

- two all-round lights in a vertical line, the upper being green and the lower white.

Memory Aid: *"Green over white, nets are in sight."*

Fishing/trawling vessel, we replace the masthead with:

- two all around lights, red or green (depending on vessel type) over a white all around.
- Use day shape of two cones with apexes together in a vertical line, one above the other.
- When outlying gear extends more than 150 meters horizontally from the vessel, display a cone (apex upward) in the direction of the gear and (at night) a white light.
- Signals for trawlers:
 - When shooting their nets: two white lights in a vertical line;
 - When hauling their nets: one white light over one red light in a vertical line;
 - When a net has come fast upon an obstruction: two red lights in a vertical line.
- Each vessel engaged in pair trawling may exhibit by night, a searchlight directed forward and in the direction of the other vessel of the pair;
- Signals for purse seiners. Vessels engaged in fishing with purse seine gear may exhibit two yellow lights in a vertical line. These lights shall flash alternately every second and with equal light and occultation duration. These lights may be exhibited only when the vessel is hampered by its fishing gear.

RULE 27: Vessels Not Under Command

Vessel not under command is not able to maneuver due to special circumstances (e.g., boat is broken). It must exhibit:

- two all-round red lights in a vertical line (where they can best be seen).
- *When making way*, it also must display sidelights and a sternlight, but no masthead light.

Memory Aid: “Red over red, the Captain’s dead.”

Dayshape: Two balls in a vertical line

Vessel Restricted in Ability to Maneuver is doing work dredging, cable laying, surveying, transferring goods and people when making way, performing dive operations, etc. It does NOT normally include towing operations. These are boats doing work on the water versus transporting people or cargo. They can not easily get out of the way of others. Therefore, they must display:

- three all-round lights in a vertical line (Red-White-Red), where they may best be seen.

Memory Aid: “Red-White-Red over a working man’s head.”

When making way, they display:

- sidelights,
- sternlight,
- and masthead light.
- When anchored, add anchor light(s).

Light Change from standard configuration? *Add three lights, Red-White-Red, where they can best be seen.*

Dayshape: Ball-Diamond-Ball

Normally a towing vessel is not considered restricted in its ability to maneuver. If towing operations severely restrict the towing vessel and tow in their ability to deviate from their course, they must display the required lights and shapes for being restricted in their ability to maneuver.

Vessel engaged in dredging or underwater operations have discharge lines and other obstructions to navigation. To warn others of these dangers they must also exhibit:

- two all-round red lights/two balls in a vertical line to indicate the side on which the obstruction exists.
- two all-round green lights or two diamonds in a vertical line for the passing side.

Underwater operations are restricted in ability to maneuver; the vessel cannot leave with divers down.

Day Shape: Rigid International Code flag “Alfa” may be used if the vessel can not display ball-diamond-ball. States may require the red background and white diagonal stripe, the traditional diver’s flag.



Mine clearance vessels are dangerous for others to approach within 1,000 meters. They shall exhibit (in addition to lights of a power-driven vessel):

- three all-round green lights or three balls. One shall be exhibited near the foremast head and one at each end of the fore yard.

Memory Aid: “Green-Green-Green Extreme”

RULE 28: Vessels Constrained By Their Draft (*International Rules Only*)

Vessel constrained by her draft displays lights of a power-driven vessel as required by her length, and may also display (where they can best be seen):

- three all-round red lights in a vertical line.

Memory Aid: “Rudders Rubbing Rock”

Light Change from standard configuration? *We have added three all-round red lights, where they can best be seen.*

Dayshape: a cylinder (**Memory Aid:** Think draft and cylinder like draft beer.)

RULE 29 Pilot Vessel

When on duty, a pilot vessel must exhibit at or near the masthead:

- two all-round lights in a vertical line, the upper white and lower red.

When underway, a pilot vessel must exhibit:

- sidelights and sternlight.

When not on duty, it complies with lights or day shapes rules for a power-driven vessel of her length.

Memory Aid: “White over Red, pilots making bread (money).”

RULE 30: Anchored and Aground Vessel

For vessels that are NOT underway:

- vessel less than 50 meters in length at anchor shall exhibit, where it can best be seen, all-round white light.

Dayshape: For any size vessel is a single black ball

A vessel 50 meters or more in length at anchor shall exhibit (where it can best be seen):

- in the fore part, an all-round white light or one ball,
- an all-round white light (at or near the stern, and at a lower level than the light in front).

A vessel of 100 meters or more in length at anchor shall also use working lights to illuminate her decks.

Think of it as pointing from the front to back when it is safe to pass.

A vessel aground shall exhibit:

- the required anchor lights, and in addition, where they can best be seen,
- two all-round red lights in a vertical line, and
- three balls in a vertical line. (One ball from “*at anchor*,” plus two balls from “*not under command*” equal three balls.)

EXCEPTIONS:

- Vessel < 7 meters (23 ft) in length, when at anchor and not in/near a narrow channel, fairway, anchorage, or where other vessels normally navigate, is not required to exhibit these lights or shapes.
- Vessel < 12 meters (39.6 ft) in length when aground is not required to exhibit required lights or shapes.
- Vessel < 20 meters and anchored in special anchorage area designated by the Secretary is not required to exhibit anchor lights/shapes required by Inland Rules.

RULE 31: Seaplanes

It may not be practicable for a seaplane to exhibit lights and shapes as required in the Rules. She must exhibit lights and shapes as closely as is possible.

RULE 32: Definitions of Sound and Light Signals

Whistle means any sound signaling appliance capable of producing the prescribed blasts, and which complies with specifications in Annex III to these Rules.

Short blast means a blast of about one second duration.

Prolonged blast means a blast of 4 to 6 second duration.

RULE 33: Equipment For Sound Signals (Inland/International)

- A vessel < 12 meters (39.5 ft) in length is not obliged to carry the sound signaling appliances, but shall be provided with some other means of **making an efficient sound signal**.
- A vessel 12 meters to 20 meters (65 ft) in length shall be provided with a **whistle**.
- A vessel 20 meters or more in length shall be provided with a **whistle and a bell**.
- A vessel 100 meters (328 ft) or more in length must have a **gong**, which cannot be confused with a bell.

RULE 34: Maneuvering—Warning Signals

(*Inland Rules*) Required of power-driven vessels in sight of one another within half a mile of each other, and underway in a **meeting or crossing situation**, signal your intention by:

- One short blast = “*I intend to **leave you on my port side.***”
- Two short blasts = “*I intend to **leave you on my starboard side.***”
- Three short blasts = “*I am operating astern propulsion.*”

Hearing one or two blast signals and agreement, sound the same signal and pass safely. If in doubt/danger, signal **five or more short blasts**.

Required of power-driven vessels in sight of one another within half a mile of each other, and underway in a **overtaking situation**, signal your intention by:

- One short blast = “*I intend to overtake you on **your** starboard side*” (or **my port**).
- Two short blasts = “*I intend to overtake you on **your** port side*” (or **my starboard**).

A power-driven vessel being **overtaken in agreement**, sounds a similar sound signal. If in doubt/danger, **five or more short blasts** shall be used.

NOTE: No matter how it is stated, if you are showing your port to the other boat, it is **one short my port**.

If you reach an agreement with another vessel by radiotelephone (as prescribed) in such situations as a head-on, crossing, or overtaking, you are not obliged to sound the whistle signals prescribed by this Rule, but may do so. (*Inland Rules Only.*) International rules require you to sound whistle signals.

(*International Rules*) When vessels are in sight of one another (“*When I see you*”), a **power-driven vessel underway** shall signal her intentions with:

- One short blast = “*I am altering my course to starboard.*”
- Two short blasts = “*I am altering my course to port.*”
- Three short blasts = “*I am operating astern propulsion.*”

The other vessel is not obligated to respond. The danger/doubt signal may be used, but is not required.

Required of power-driven vessels in sight and within half a mile of each other, and underway. Under **International Rules**, when **in sight of one another in a narrow channel/fairway** a vessel intending to overtake another shall indicate intention by:

- Two prolonged blasts followed by one short blast = “*I intend to overtake you on your starboard side.*”
- Two prolonged blasts followed by two short blasts = “*I intend to overtake you on your port side.*”

The vessel about to be overtaken shall indicate her agreement with **one prolonged, one short, one prolonged and one short blast**, in that order.

Required of power-driven vessels in sight of one another within half a mile of each other, and underway. A **vessel is nearing a bend, channel, or fairway where other vessels may be obscured by any obstruction** shall indicate intention by:

- sounding one prolonged blast.

This signal shall be answered with a prolonged blast by any approaching vessel that may be within hearing around the bend or obstruction.

Passing signals are still needed when in sight of each other.

A **power-driven vessel is leaving a dock or berth**. She shall sound:

- one prolonged blast. (*Inland Rules Only*)

RULE 35: Sound Signals In Restricted Visibility

If you are in or near an area of restricted visibility, whether by day or night, the following signals must be used to avoid a collision.

Power-driven vessel making way shall sound at intervals not more than **two** minutes:

- one prolonged blast.

Power-driven vessel underway but stopped/not making way shall sound every two minutes (not broken):

- two prolonged blasts.

All others below shall sound every **two** minutes one prolonged followed by two short blasts.

- Vessel not under command
- Vessel restricted in her ability to maneuver, whether underway or at anchor
- Vessel constrained by draft
- Vessel engaged in fishing, whether underway or at anchor
- Sailing vessel
- Vessel engaged in towing or pushing another vessel
- ***Every vessel that is not just power driven!***

If the **last towed vessel is manned**, it shall at intervals of not more than **two** minutes sound **one prolonged followed by three short blasts** IMMEDIATELY after the signal made by the towing vessel.

Pilot vessel on duty *may* sound four additional short blasts after the required signals.

For **vessels NOT UNDERWAY but away from docks**, you need to let others know where you are.

- A vessel at anchor shall every **one** minute ring the bell rapidly for about 5 seconds.
- A vessel of 100 meters (328 ft) or more in length shall sound a gong rapidly for 5 seconds in the after part of the vessel, immediately after the ringing of the bell.
- A vessel (of any size) at anchor may in addition sound **one short, one prolonged and one short blast**, to give warning of her position.
- Vessel aground shall **give the bell** and if required, the gong signal, **then must give three separate and distinct strokes** on the bell immediately before and after the rapid ringing of the bell.
- Vessel < 12 meters (39.4 ft) is not obliged to give the anchor/aground signals but must make **some other efficient sound signal** at intervals of not more than **two** minutes apart.

Vessels anchored in a special anchorage area designated by the Secretary **shall not be required** to sound signals if < 20 meters (65.6 ft) in length.

RULE 36: Signals To Attract Attention

If you need to attract the attention of another vessel, use light or sound signals that **cannot** be mistaken for any signal authorized elsewhere in these Rules. You can direct a searchlight in the direction of the danger, ***“in such a way as not to embarrass any vessel.”***

Embarrass (archaic) means to hamper or impede (a person, movement, or action).

Easy Reminders:

- Maneuvering - Start with a short
- Restricted visibility (fog, around a bend, leaving the dock) - Sound a prolong
- At anchor - Always start with a bell

RULE 37: Distress Signals

DEFINED IN ANNEX IV

The following signals, used or exhibited either together or separately, indicate distress and need of assistance:

- a gun or other explosive signal fired at intervals of about a minute
- a continuous sounding with any fog-signaling apparatus
- rockets or shells, throwing red stars fired one at a time at short intervals
- a signal made by any signaling method consisting of the group. . - - - . . (SOS) in the Morse Code
- a signal sent by radiotelephony consisting of the spoken word "Mayday"
- the International Code Signal of distress indicated by November Charlie.
- a signal consisting of a square flag having above or below it a ball or anything resembling a ball
- flames on the vessel (as from a burning tar barrel, oil barrel, etc.)
- a rocket parachute flare or a hand flare showing a red light
- a smoke signal giving off orange-colored smoke
- slowly and repeatedly raising and lowering arms outstretched to each side
- a distress alert by means of digital selective calling (DSC) transmitted on:
 - (i) VHF channel 70, or
 - (ii) MF/HF on the frequencies 2187.5 kHz, 8414.5 kHz, 4207.5 kHz, 6312 kHz, 12577 kHz or 16804.5 kHz
- a ship-to-shore distress alert transmitted by the ship's Inmarsat or other mobile satellite service provider ship earth station
- signals transmitted by emergency position-indicating radio beacons
- approved signals transmitted by radio communication systems, including survival craft radar transponders

Signals used together or separately indicate distress and need of assistance. Under the ***Good Samaritan Law*** you are required to provide assistance.

A high intensity white light flashing at regular intervals from 50 to 70 times per minute may also be used (*Inland Rules Only*).

RULE 38: EXEMPTIONS

As with any code or law there are exemptions. There were a number of changes in the U.S. in the 1980's. Any vessel or class of vessels, the keel of which is laid, or which is at a corresponding stage of construction before December 24, 1980, provided that she complies with the requirements of:

- (a) The Act of June 7, 1897 (30 Stat. 96), as amended (33 U.S.C. 154- 232) for vessels navigating the waters subject to that statute;
- (b) Section 4233 of the Revised Statutes (33 U.S.C. 301-356) for vessels navigating the waters subject to that statute;
- (c) The Act of February 8, 1895 (28 Stat. 645), as amended (33 U.S.C. 241-295) for vessels navigating the waters subject to that statute; or
- (d) Sections 3, 4, and 5 of the Act of April 25, 1940 (54 Stat. 163), as amended (46 U.S.C. 526 b, c, and d) for motorboats navigating the waters subject to that statute;

shall be *exempted* from compliance with the technical Annexes to these Rules as follows:

- (i) the installation of lights with ranges;
- (ii) the installation of lights with color specifications as prescribed in Annex I to these Rules.

There are a number of annexes with rules. These are used to explain in more detail the meaning of portions of the rules such as lights and sound signals. These include:

ANNEX I, 33 CFR 84: Positioning and Technical Details of Lights and Shapes

ANNEX II, 33 CFR 85: Additional Signals for Fishing Vessels Fishing in Close Proximity

ANNEX III, 33 CFR 86: Technical Details of Sound Signal Appliances.

ANNEX IV, 33 CFR 87: Distress Signals

We are just going to talk about **ANNEX V, 33 CFR 88:** Pilot Rules (*Inland Rules*).

§ 88.09 Temporary exemption from light and shape requirements when operating under bridges. A vessel's navigation lights and shapes may be lowered if necessary to pass under a bridge.

§ 88.05 After Jan. 1, 1983, operator of each self-propelled vessel 12 meters or more in length shall carry on board and maintain for ready reference a *copy of the Inland Navigation Rules*. (Now must be readily available)

§ 88.11 Law enforcement vessels. Law enforcement vessels may display a *flashing blue light* when engaged in direct law enforcement or public safety activities. This light must be located so that it does not interfere with the visibility of the vessel's navigation lights.

§ 88.12 Public safety activities. Vessels engaged in government sanctioned public safety activities, and commercial vessels performing similar functions, may display an *alternately flashing red and yellow light signal*. Light signal may be used only as an identification signal and *conveys no special privilege*.

§ 88.15 Lights on dredge pipelines. Dredge pipelines floating or supported on trestles shall display the following lights at night and in periods of restricted visibility: *one row of yellow lights that* must be:

- Flashing 50 to 70 times per minute
- Visible all around the horizon, at least 2 miles on a clear dark night
- From 1 to 3.5 meters above the water, and equally spaced and sufficient in number to clearly show the pipeline's length and course
- *Two red lights* must be displayed at each end of the pipeline, including the ends in a channel where the pipeline is separated to allow vessels to pass (whether open or closed). The lights must be one meter apart in a vertical line with the lower light at the same height above the water as the flashing yellow light.

With a **Dredge Operation** you can see the following:

- Two red lights or two black balls to one side of the dredge which indicate *do not pass, unsafe*
- Two green lights or two black diamonds, *pass this side*
- Red-White-Red, or Ball-Diamond-Ball, *restricted vessel*
- Row of yellow lights, *pipeline*
- Two sets of black balls or red lights in vertical line, *pass through pipeline*

SECTION 2: MEMORY AIDS

Memory Aids: Memorize these to help you remember light configurations

“Red over Red, the Captain is dead” (Not Under Command)

“Red-White-Red, over working man’s head” (Restricted in Ability to Maneuver)

“Rudder Rubbing Rocks” (Constrained by Draft) (*International Rules*)

“Red over White, fishing tonight” (Fishing Vessel)

“Green over White, trawling tonight” or *“nets are in sight”* (Trawler)

“Red over Green, I am a sailing machine” (Sailboat Under Sail)

“White over Red, pilots making bread” (Pilot Vessel on Duty)

“Yellow over White, we are not alone tonight” (Towing Astern)

“White over White, I am towing tonight” (Master Heads on a Towing Vessel)

“Yellow over Yellow, you are a Pushy Fellow” (Push or Hip Tow) (*Inland Rules Only*)

“Watch out for the FLASHING BLUE LIGHT special” (Enforcement)

“Green-Green-Green, Danger Extreme” (Mine Sweeper)

“P-O-W have no rights” (Port - Overtaking - Windward are Giveaway)

Memory Aid for this list of privileged vessels is:

<u>New</u>	Not Under Command (Two red lights; Two black balls; One prolong, two shorts)
<u>Reels</u>	Restricted in Ability to Maneuver (Red-white-red; Ball-diamond-ball; One prolong, two shorts)
<u>Catch</u>	Constrained by Draft (Three red lights; Cylinder; One prolong, two shorts) (<i>International Rules Only</i>)
<u>Fish</u>	Fish or Trawler, Commercial (Red over white; Green over white; Two cones apex to apex; One prolong, two shorts)
<u>So</u>	Sail Only (Not Under Power) (Red over green, opt.; No day shapes; One prolong, two shorts)
<u>Purchase</u>	Power-Driven (No day shapes, except inverted cone for power sailing)
<u>Some</u>	Seaplane (Lights - best as can be done to comply with Rules; No day shapes; One or two prolongs)

In class I talk about *special vessels*. Why are they special, they have lights, sounds and dayshapes to indicate special privileges. From Not Under Command to Sailing, each vessel has the same restricted sound signal, special lights or lack of lights, and dayshapes to tell you who they are. It is just like the highway crew working on the roads. This makes it easier to remember lights and sounds.

Standard Light Configuration:

- Sidelights, mast, and stern
- Remember, a second masthead light abaft and higher is a big boat (50 Meters or more)

Remember: only vessels involved with towing have **yellow lights**

Other Helpful Memory Aids:

I am a dumbship, dumbship, dumbship, as I ran my ship aground. (Three balls in a vertical line)

"One bright white light, I am anchored tonight." (< 50 Meters)

"Two bright white lights, my ends are out of sight." (At anchor, greater than or equal to 50 Meters)

Sound Signal Memory Aid: In Restricted Visibility: Underway Make a list from 1 to 5.

Number represents the number of sound signals. **First one is a prolong**

- 1 - One prolong (Moving) Making Way, Power-Driven,
 2 - Two prolongs (NOT Moving) Underway, No-Way, Power-Driven
 3 - One prolong and two shorts (A L L) All Special Vessels
 4 - One prolong and three shorts (Manned T O W) Manned Tow, if last tow
 5 - One prolongs and four shorts (P I L O T) Pilot Vessel
 One prolong Leaving a Dock (*Inland Rules*) or Approaching an Obstructed Bend

Maneuvering: In sight of another (Inland)

- One short I intend to leave you on my port
 Two short I intend to leave you on my starboard
 Three short I am operating in stern propulsion
 Five short or more Danger/doubt

Maneuvering: In sight of another (International)

- One short I am altering course to starboard
 Two short I am altering course to port
 Three short I am operating in stern propulsion
 Five short or more Danger/doubt

When overtake another shall indicate intention by:

Two prolonged blasts then one short blast = I intend to overtake you on your starboard side.

Two prolonged blasts then two short blasts = I intend to overtake you on your port side.

Vessel overtaken agrees = one prolonged, one short, one prolonged and one short blast.

In Restricted Visibility: Not underway (NOTE: All start with a bell; opt. happens after the bell)

Five seconds of a bell At anchor < 100 M
 Five seconds of a gong Added if greater than or equal to 100 M
 Three separate distinct strokes (before and after bell) Aground
 One short, one prolonged, one short whistle Optional at anchor

REMEMBER - Here are several ways to remember Inland Rules

One Short, I am showing you my port. Shorts are used only when I see you (Shorts on a sunny day or warm clear night).

Prolongs are loud and long so I can hear you in the fog, around a bend, and leaving the dock).

Bells are used when anchored or aground to let others know you are attached to the ground.

Three strikes of a bell or three balls means "I am a dumb ship, dumbship dumbship as I ran my ship aground".

Power driven is a generic vessel. In the fog it sounds either one or two prolongs. All other vessels have "**Special**" names and a special fog signal - One prolong and two shorts (Not Under Command, Restricted in the Ability to Maneuver, Constraint by Draft, Fishing, Trawling, Sailing, and Towing.)

Pilot will sound one or two prolongs of a power driven vessel and when she wants to be found add four shorts.

Manned Tow, if at the end, will sound one prolong and three shorts.

Must Remember: **Short** I see you; **Prolong** I do NOT see you; **Bell** not underway

Rules Practice Exam 1-19

1. Complete the definitions (One answer per space)

1. The purpose of these Rules is to ____ a) vessel not anchor, moored, or aground
 2. Navigation Rules consist of International and Inland Rules separated by the ____ b) commercial fishing
 3. ____ applies to all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels. c) International Rules
 4. ____ and ____ shall not impede the passage of vessel navigating within a narrow channel by ____, ____, or _____. d) Risk of collision exists
 5. A power-driven vessel, down bound with a following current on western rivers, has the _____. e) vessel maneuverability
 6. A power-driven vessel, operating in narrow channels on Western Rivers, proceeding down bound with a following current meeting an upbound vessels shall: ____ and _____. f) Initiate maneuvering signals as appropriate.
 7. We depart from the rules to avoid ____ g) crossing which may impede
 8. Vessel restricted in her ability to maneuver by the nature of her work, includes ____ h) anchoring in a narrow channel.
 9. Underway means a _____. i) Propose the manner and place of passage
 10. ____ by sight and hearing, by all available means appropriate in the prevailing circumstances and conditions of the risk of collision. j) immediate danger.
 11. To allow more time to assess the situation, _____. k) transferring persons or cargo underway
 12. Safe Speed, so you can _____. l) Sailing vessel
 13. ____ if the compass bearing of an approaching vessel does not appreciably change. m) Proper look-out
 14. ____ shall be positive, made in ample time, and, with due regard to the observance of good seamanship. n) Avoid collisions.
 15. Criteria to determine Safe Speed includes ____ and _____. o) Action taken to avoid collision
- p) COLREGS Demarcation Lines.
- q) state of the wind, sea, and current
- r) right-of-way over an upbound vessel,
- s) Vessel less than 20 meters
- t) a vessel shall slacken her speed
- u) ...take proper and effective action to avoid collision, and be stopped within a distance appropriate to the prevailing circumstance and conditions

2. Open water crossing/head-on situation, prioritize vessels with (1) being the highest priority to (6) being the lowest

- ___ Vessel not under command
- ___ Vessel engaged in fishing
- ___ Sailing vessel underway
- ___ Vessel restricted in her ability to move
- ___ Power driven tow
- ___ Vessel engaged in trawling
- ___ Sailing vessel motoring
- ___ Vessel constrained by her draft

3. Determine the following vessels status

- 1) Any vessel overtaking another is always ____
- 2) A sailboat gives way to a power vessel when it's ____
- 3) Power driven vessels give way to another power driven vessel on their ____ side
- 4) Power drive vessels in a head-on situation must go to ____
- 5) Two meeting sailboats, the one to ____ or ____ is give way

- a. Windward
- b. Overtaking
- c. Starboard
- d. Port
- e. Give-way
- f. Stand-on

Rules Practice Exam 20 - 31

1. Match the light characteristic with the specific vessel light.

- | | |
|--|-------------------|
| 1) Sidelights on the port side are _____ | a. Sternlight |
| 2) A white light over the vessel centerline, facing forward, is the _____ | b. Masthead light |
| 3) The Masthead Light shows an arc of light _____ degrees over the horizon | c. 180 |
| 4) A white light on the stern is the _____ | d. White |
| 5) Sidelights on the starboard side are _____ | e. Towing light |
| 6) Sternlight shows an arc of light _____ degrees over the horizon | f. Blue |
| 7) The _____ is yellow and has the same characteristics of the Sternlight | g. Red |
| | h. 225 |
| | i. Green |
| | j. 135 |
| | k. 112.5 |
| | l. Yellow |

2. Fill in the blank..... "Overtaking is when the stern light can be seen _____ degrees"

3. Circle True or False? "When towing on the hip, inland, you must exhibit a sternlight above the towing light on the tug."

4. Lights help identify different vessels. Match the correct vessel for each set of lights.

- | | |
|--|--|
| 1) Two all-round lights; upper red and lower white, sidelights and sternlight _____ | a. Towing on the hip, Inland |
| 2) Two all-round lights; upper red and lower green, sidelights, and sternlight _____ | b. Power-driven vessel under 50 M, under-way |
| 3) Two towing lights in vertical line, no masthead, and sidelights _____ | c. Vessel engaged in fishing |
| 4) Two masthead lights in vertical line, sidelights, and towing light in vertical line above sternlight. _____ | d. Power-driven vessel towing astern |
| 5) Two masthead lights in a vertical line, sidelights, and two towing lights in a vertical line. _____ | e. Sailing vessel |
| 6) Masthead light, sidelights, and sternlight _____ | f. Power-driven vessel pushing ahead above the Huey P. Long Bridge |
| | g. Power-driven vessel 50 M or more, under-way |

5. Match the correct day shapes with their meaning.

- | | |
|--|---|
| 1) Restricted in her ability to maneuver _____ | a. Diamond |
| 2) At anchor _____ | b. Three balls in a vertical line |
| 3) Towing astern over 200 M _____ | c. Cone, apex up |
| 4) Not under command _____ | d. Two Diamonds in a vertical line |
| 5) Aground _____ | e. Two balls in a vertical line |
| 6) Fishing/Trawling _____ | f. Two sets of two balls, each set in a vertical line |
| 7) Do not pass on this side _____ | g. Ball, Diamond, Ball in a vertical line |
| 8) Constraint by draft _____ | h. Cylinder |
| 9) Fishing gear out 150 M _____ | i. Ball |
| 10) Pipeline Opening _____ | j. Two cones, apex to apex |
| 11) Motor Sailing _____ | k. Two balls at each end of vessel |
| 12) Pass this side of obstruction _____ | l. Cone, apex down |

6. Match the length of the vessel with the proper display of lights **when at anchor**.

- | | |
|-------------------------------------|---|
| 1) Less than 50 M _____ | a. Deck Lights |
| 2) Over 100 M _____ | b. Two all-round white lights with one abaft and lower |
| 3) Over 50 M, less than 100 M _____ | c. One all-around green light |
| 4) Under 7 M _____ | d. One all-round white light |
| | e. None |
| | f. Two all-round white lights with one abaft and higher |

Rules Practice Exam 32 -36

1. Match the definition with the sound

- | | |
|---|------------------------------|
| 1) I intend to leave you on my port side _____ | a. Three short blasts |
| 2) I am operating astern propulsion _____ | b. One short blast |
| 3) I intend to leave you on my starboard side _____ | c. Five or more short blasts |
| 4) If in doubt/danger, signal _____ | d. Four Short Blast |
| | e. Two short blasts |

2. Choose all of the correct answers for the following question: "A vessel of 100 meters or more, NOT underway (anchored) but AWAY from dock, shall sound....."

- _____ 1. Ring the bell rapidly for 5 seconds every 5 minutes
- _____ 2. Ring the bell rapidly for 5 seconds every minute
- _____ 3. Sound the gong once immediately after ringing the bell
- _____ 4. Sound the gong rapidly immediately after ringing the bell

3. Match the length of the vessel with the sound making device(s), they must carry.

- | | |
|--|----------------------------------|
| 1) Vessel 12 - 20 M _____ | a. Bell |
| 2) Vessel 20 M and more _____ | b. Whistle |
| 3) Vessel 100 M or more _____ | c. Gong |
| 4) Vessel less 15 M in special anchorage _____ | d. None |
| 5) Vessel less than 12 M _____ | e. Effective sound making device |

4. Fill in the blanks..... "A prolonged blast is a blast in duration of ____ to ____ seconds"

5. Circle True or False... "A vessel aground shall give the bell and gong, if required, then four separate and distinctive strokes of the bell before and after the rapid ringing of the bell"

6. Choose all of the correct answers about a vessel at anchor in restricted visibility

- _____ 1. Shall rapidly ring bell for five seconds every two minutes
- _____ 2. May also sound one short, one prolonged, one short blast
- _____ 3. Shall ring bell rapidly for five seconds every minute
- _____ 4. If under 12 meters, can give any efficient sound signal every two minutes
- _____ 5. If under 12 meters, can give any efficient sound signal every one minute

Rules Practice Exam 37 - 38

1. Match the lights or dayshape to its meaning when encountering a **dredge operation**.

- | | |
|---|---|
| 1) Two red lights or two black balls to one side of the dredge_____ | a. Pass this side |
| 2) Two green lights or two black diamonds_____ | b. Underway |
| 3) Red-White-Red or Ball-Diamond-Ball_____ | c. Pipeline |
| 4) Row of Yellow Lights_____ | d. Vessel Restricted in ability to maneuver |
| 5) Two sets of black balls or red lights in a vertical line_____ | e. Do not pass, unsafe |
| | f. Pass through pipeline |
| | g. Not underway |

2. Choose all of the correct signals that may indicate distress and need of assistance.

- _____ 1. Flames on vessel deck
- _____ 2. A square flag with a cone, apex down
- _____ 3. Orange smoke signals
- _____ 4. A rocket parachute with a red light
- _____ 5. Jumping up and down yelling
- _____ 6. A rocket parachute with a white light
- _____ 7. Five or more short blast

3. Fill in the blank..... "Law enforcement vessels may display a flashing _____ light when engaging in law enforcement activities."

4. Circle True or False... "A gun or explosive signal fired about every minute signals distress."

RULES 1-19

Practice 1

- 1) n
- 2) p
- 3) c
- 4) s, l and g b h
- 5) r
- 6) i, f
- 7) j
- 8) k
- 9) a
- 10) m
- 11) t
- 12) u
- 13) d
- 14) o
- 15) e, q

Practice 2

- 1
- 4
- 5
- 2
- 6
- 4
- 6
- 3

Practice 3

1. b
2. b
3. c
4. c
5. d, a

Rules 20-31

Practice 1

- 1) g
- 2) b
- 3) h
- 4) a
- 5) i
- 6) j
- 7) e

Practice 2

135

Practice 3

False

Practice 4

- 1) c
- 2) e
- 3) f
- 4) d
- 5) a
- 6) b

Practice 5

- 1) g
- 2) i
- 3) a
- 4) e
- 5) b
- 6) j
- 7) e
- 8) h
- 9) c
- 10) f
- 11) l
- 12) d

Practice 6

- 1) d
- 2) a, b
- 3) b
- 4) e

RULES 32-36

Practice 1

- 1) b
- 2) a
- 3) e
- 4) c

Question 2

- 2
- 4

Question 3

- 1) b
- 2) a, b
- 3) a, b, c
- 4) d
- 5) e

Question 4

4-6

Questions 5

False - 3 separate strokes

Question 6

- 2
- 3
- 4

RULES 37 - Appendix 5

Practical 1

- 1) e
- 2) a
- 3) d
- 4) c
- 5) f

Practical 2

- 1
- 3
- 4

NOTE: not #5

Practical 3

Blue

Practical 4

True